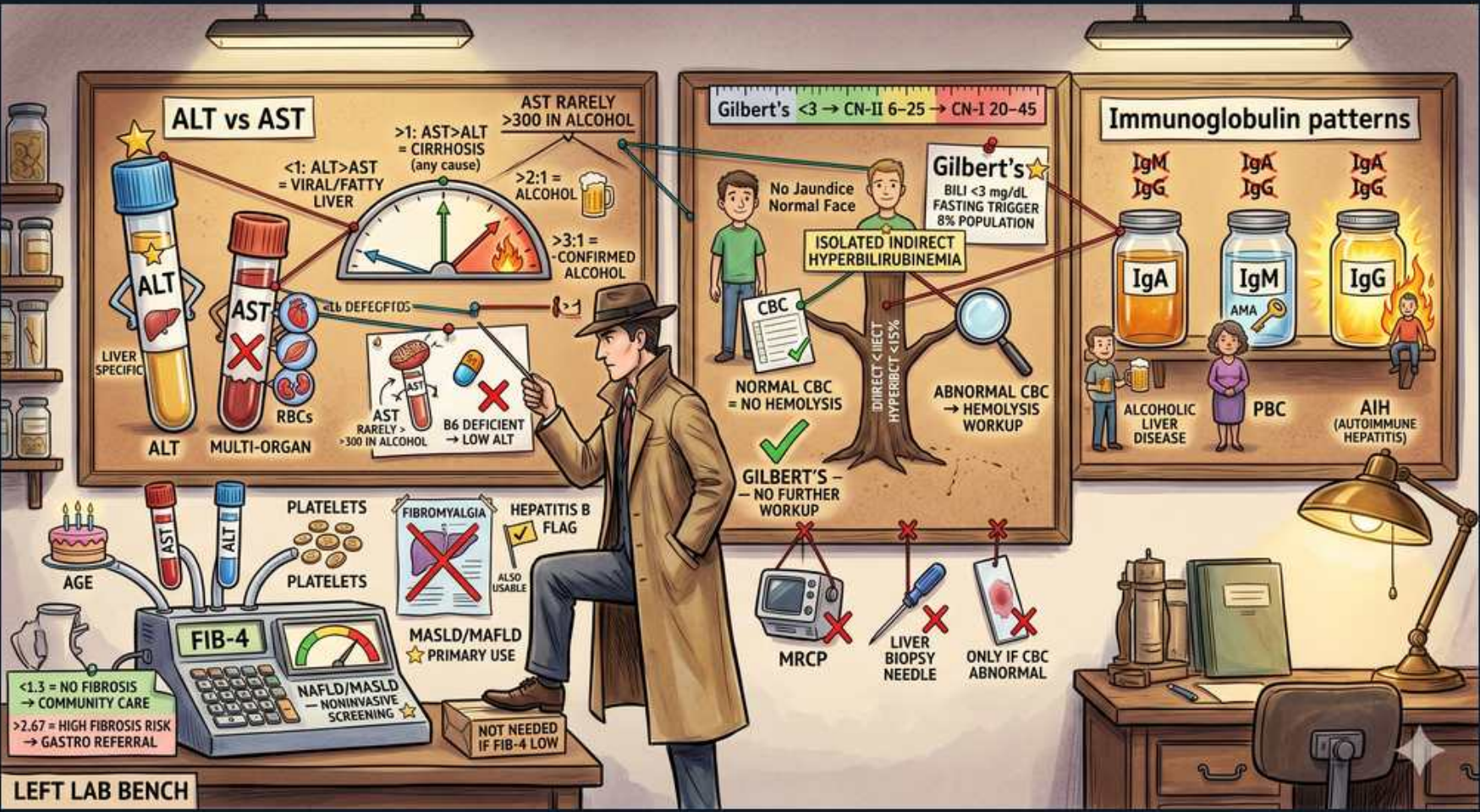


Liver Approach — Left Lab Bench



1. ALT tube (liver-specific)

WHAT YOU SEE

Tall tube labeled ALT, 'LIVER SPECIFIC'

WHAT IT ENCODES

ALT (alanine aminotransferase) is the LIVER-specific transaminase — it is concentrated in hepatocytes, so its elevation reliably localizes injury to the liver. An ALT-predominant (ALT > AST) hepatocellular pattern classically reflects viral hepatitis or NAFLD/MASLD (fatty liver). In chronic HCV and steatohepatitis ALT can be only mildly elevated yet still mark ongoing injury, so it anchors the noninvasive fibrosis workup.

BOARD TRAP

WRONG answer: assuming a normal ALT excludes significant liver disease/fibrosis. Discriminator: ALT can be normal in advanced fibrosis/cirrhosis and in NAFLD — staging relies on FIB-4 and elastography, not on the ALT value alone.

2. AST tube (multi-organ)

WHAT YOU SEE

Blood tubes labeled AST, 'MULTI-ORGAN'

WHAT IT ENCODES

AST (aspartate aminotransferase) is MULTI-organ: it comes from liver, skeletal muscle, cardiac muscle, kidney and red blood cells, so it is NOT liver-specific. An isolated AST rise should prompt a non-hepatic search (CK for muscle/rhabdomyolysis, hemolysis, recent MI, strenuous exercise) before blaming the liver.

BOARD TRAP

WRONG answer: diagnosing liver disease from an isolated AST elevation with a normal ALT. Discriminator: check CK and a hemolysis panel — muscle injury (rhabdo, myositis, macro-AST) or hemolysis raises AST without liver disease; a truly hepatic process usually moves ALT too.

3. AST:ALT dial >2:1

WHAT YOU SEE

Dial showing ratios: >1 viral/fatty, >2:1 alcohol, >3:1 confirmed alcohol

WHAT IT ENCODES

The AST:ALT (De Ritis) ratio triages hepatocellular injury: ALT > AST (ratio <1) suggests viral hepatitis or NAFLD; AST:ALT >2:1 suggests alcoholic liver disease; >3:1 strongly supports established ALD. A rising ratio toward/above 1 in known NAFLD also signals progression to advanced fibrosis/cirrhosis, and any cause plus a low albumin/high INR signals cirrhosis.

BOARD TRAP

WRONG answer: using an AST:ALT >2:1 to call viral hepatitis or to diagnose ALD by ratio alone. Discriminator: viral/NAFLD give ALT>AST (ratio <1); the >2:1 pattern points to alcohol, but NAFLD that has progressed to cirrhosis can also flip the ratio above 1 — interpret with the clinical context, not the number in isolation.

4. AST>300 in alcohol rare

WHAT YOU SEE

'AST rarely >300 in alcohol'

WHAT IT ENCODES

In pure alcoholic hepatitis the AST rarely exceeds ~300-400 IU/L despite a sick patient; the injury is ratio-defined (AST:ALT >2) rather than magnitude-defined. The relatively modest AST cap is a key board discriminator from the >1000 causes.

BOARD TRAP

WRONG answer: attributing transaminases in the thousands to alcohol. Discriminator: AST/ALT >1000 are NOT alcohol — think ischemic ('shock') hepatitis, acute viral hepatitis, or toxin (acetaminophen); reconsider the diagnosis when an apparent drinker has very high values (look for coingested acetaminophen, hypotension, viral cause).

5. B6 deficiency / low ALT

WHAT YOU SEE

'B6 deficiency low ALT in alcohol' note

WHAT IT ENCODES

ALT synthesis requires pyridoxal-5'-phosphate (vitamin B6). Chronic alcoholics are B6-deficient, so they make less ALT — this is the mechanism that DROPS the ALT and raises the AST:ALT ratio above 2 in alcoholic liver disease (it is the AST:ALT ratio's underlying explanation, not just a footnote).

BOARD TRAP

WRONG answer: interpreting the low ALT in an alcoholic as evidence of mild/improving liver disease. Discriminator: the low ALT is a B6-deficiency artifact inflating the ratio — it does not mean less injury; severity is judged by Maddrey's discriminant function/MELD, not by the falsely low ALT.

6. Gilbert range bar

WHAT YOU SEE

Bar: Gilbert <3, CN-II 6-25, CN-I 20-45

WHAT IT ENCODES

Unconjugated hyperbilirubinemia from impaired UGT1A1 conjugation grades by bilirubin level: Gilbert syndrome typically <3 mg/dL (reduced UGT1A1 activity), Crigler-Najjar type II ~6-25 mg/dL (severe partial deficiency, responds to phenobarbital), Crigler-Najjar type I ~20-45 mg/dL (near-absent enzyme, kernicterus risk, needs phototherapy/transplant).

BOARD TRAP

WRONG answer: treating a mildly elevated INDIRECT bilirubin (<3) as a cholestatic/obstructive problem warranting imaging. Discriminator: isolated unconjugated hyperbilirubinemia with normal ALP/transaminases and no hemolysis is Gilbert — benign, no imaging or biopsy; cholestasis is CONJUGATED with high ALP.

7. Gilbert no jaundice face

WHAT YOU SEE

Calm face, 'No jaundice / normal face'

WHAT IT ENCODES

Gilbert syndrome is a benign, common (~5-10%) inherited reduction in UGT1A1 conjugation producing mild, fluctuating, often subclinical indirect hyperbilirubinemia. It is unmasked by FASTING, illness, stress, dehydration or exertion, and is otherwise asymptomatic with a completely normal exam — no treatment needed.

BOARD TRAP

WRONG answer: launching a hepatitis/cholestasis workup or biopsy in a young patient with intermittent mild jaundice during illness/fasting. Discriminator: normal CBC, normal LFTs except isolated indirect bilirubin, no hemolysis = Gilbert; the trigger (fasting/stress) and otherwise normal labs clinch it without further testing.

8. Normal CBC, no hemolysis

WHAT YOU SEE

'NORMAL CBC, NO HEMOLYSIS'

WHAT IT ENCODES

For isolated unconjugated hyperbilirubinemia, a NORMAL CBC with no hemolysis markers (normal reticulocytes, LDH, haptoglobin, no schistocytes) effectively confirms Gilbert and ends the workup — no imaging, no biopsy.

BOARD TRAP

WRONG answer: ordering a hemolysis panel/imaging anyway when the CBC is normal and bilirubin is isolated indirect. Discriminator: with a normal CBC and no hemolysis the cause is Gilbert by default; pursuing hemolysis labs or biliary imaging is the over-testing trap.

9. Abnormal CBC → hemolysis

WHAT YOU SEE

'ABNORMAL CBC hemolysis workup'

WHAT IT ENCODES

If the CBC is ABNORMAL (anemia) alongside unconjugated hyperbilirubinemia, pursue a hemolysis workup: reticulocyte count (high), LDH (high), haptoglobin (low), peripheral smear (schistocytes/spherocytes), and a direct Coombs/DAT. Hemolysis overproduces bilirubin faster than the liver can conjugate it.

BOARD TRAP

WRONG answer: attributing the indirect hyperbilirubinemia to Gilbert when the patient is anemic. Discriminator: anemia + reticulocytosis + high LDH + low haptoglobin = hemolysis, not Gilbert — Gilbert has a NORMAL CBC and no hemolysis markers.

10. IgG / AIH jar

WHAT YOU SEE

Jar IgG, 'AIH (autoimmune hepatitis)'

WHAT IT ENCODES

A high IgG (hypergammaglobulinemia) points to autoimmune hepatitis — a hepatocellular, interface-hepatitis disease, classically in women, with positive ANA and anti-smooth-muscle antibody (type 1) or anti-LKM-1 (type 2). Treatment is corticosteroids (prednisone) +/- azathioprine.

BOARD TRAP

WRONG answer: mistaking the high-IgG, AMA-negative cholestatic/hepatocellular picture for PBC. Discriminator: AIH = high IgG + ANA/ASMA + hepatocellular (high ALT) responding to steroids; PBC = high IgM + AMA + cholestatic (high ALP) treated with ursodeoxycholic acid — the immunoglobulin class and the LFT pattern separate them.

11. IgM / PBC jar

WHAT YOU SEE

Jar IgM with AMA, 'PBC'

WHAT IT ENCODES

A high IgM together with a positive anti-mitochondrial antibody (AMA) points to primary biliary cholangitis — autoimmune destruction of small bile ducts in a middle-aged woman, presenting with pruritus, fatigue and a cholestatic (high ALP) profile. First-line therapy is ursodeoxycholic acid 13-15 mg/kg/day.

BOARD TRAP

WRONG answer: confirming PBC by cholangiography or confusing its IgM with the IgG of AIH. Discriminator: PBC is IgM + AMA + cholestatic with a NORMAL cholangiogram (small-duct disease); AIH is IgG + ANA/ASMA + hepatocellular — the antibody and Ig class decide it, not imaging.

12. IgA alcoholic jar

WHAT YOU SEE

Jar IgA, 'ALCOHOLIC LIVER DISEASE'

WHAT IT ENCODES

A high IgA points to alcoholic liver disease (the 'A for Alcohol' mnemonic: IgA-Alcohol, IgM-PBC/Mitochondrial, IgG-autoimmune hepatitis). It complements the AST:ALT >2:1 pattern and the alcohol-related peripheral stigmata.

BOARD TRAP

WRONG answer: forgetting the Ig-to-disease mapping and pairing IgA with an autoimmune liver disease. Discriminator: IgA = alcohol, IgM = PBC, IgG = AIH — use this triad to assign the immunoglobulin to the correct etiology.

13. FIB-4 calculator

WHAT YOU SEE

FIB-4 cash-register/calculator on bench

WHAT IT ENCODES

$FIB-4 = (age \times AST) / (platelets \times \sqrt{ALT})$ — a free, four-variable index that is the recommended FIRST-LINE noninvasive fibrosis estimate in NAFLD/MASLD and other chronic liver disease. It risk-stratifies who can stay in primary care versus who needs elastography or hepatology referral, avoiding biopsy in most patients.

BOARD TRAP

WRONG answer: going straight to liver biopsy to stage fibrosis in NAFLD. Discriminator: the validated pathway is FIB-4 first, then elastography (FibroScan) for indeterminate/high values, and biopsy only when noninvasive tests are discordant or inconclusive.

14. FIB-4 <1.3 community

WHAT YOU SEE

'<1.3 low fibrosis, community care'

WHAT IT ENCODES

A FIB-4 <1.3 (or <2.0 if age >65) has a high negative predictive value for advanced fibrosis — these patients are LOW risk and can be managed in primary/community care with lifestyle modification and periodic reassessment, no specialist or biopsy needed.

BOARD TRAP

WRONG answer: referring every NAFLD patient to hepatology regardless of FIB-4. Discriminator: FIB-4 <1.3 rules out advanced fibrosis well — referral/elastography is reserved for FIB-4 >1.3 (indeterminate) or >2.67 (high risk).

15. FIB-4 >2.67 referral

WHAT YOU SEE

'>2.67 high fibrosis risk, GI/gastro referral'

WHAT IT ENCODES

A FIB-4 >2.67 indicates HIGH probability of advanced fibrosis — these patients need hepatology referral and further staging (transient elastography/FibroScan, and biopsy if discordant). Values between 1.3 and 2.67 are indeterminate and go to elastography next.

BOARD TRAP

WRONG answer: reassuring/discharging a patient with FIB-4 >2.67. Discriminator: a high FIB-4 mandates referral and elastography — treating it as low-risk misses advanced fibrosis/cirrhosis that needs varices and HCC surveillance.

16. FibroScan first

WHAT YOU SEE

FibroScan probe, 'FIBROScan noninvasive'

WHAT IT ENCODES

Transient elastography (FibroScan) measures liver stiffness in kilopascals (kPa) as a noninvasive surrogate for fibrosis stage — roughly <7 kPa low/none, ~8-12 kPa significant fibrosis, and >12-15 kPa suggestive of advanced fibrosis/cirrhosis (cutoffs vary by etiology). It is the second step after an elevated/indeterminate FIB-4 and stages fibrosis BEFORE biopsy is considered.

BOARD TRAP

WRONG answer: choosing liver biopsy as the first-line test to stage fibrosis. Discriminator: biopsy is NOT first-line — noninvasive FIB-4 then FibroScan come first; biopsy (with its bleeding risk and sampling error) is reserved for discordant or inconclusive noninvasive results.

17. Platelets low = fibrosis

WHAT YOU SEE

'PLATELETS' low flag

WHAT IT ENCODES

A falling platelet count is one of the earliest clues to advanced fibrosis/cirrhosis: portal hypertension causes splenic sequestration (hypersplenism) and reduced hepatic thrombopoietin lowers platelets. Low platelets is also why FIB-4 (which divides by platelets) rises with worsening fibrosis.

BOARD TRAP

WRONG answer: working up isolated thrombocytopenia as a primary hematologic problem (ITP, marrow) while ignoring the liver. Discriminator: in chronic liver disease, low platelets reflect portal hypertension/hypersplenism — check spleen size, FIB-4 and synthetic function before assuming a marrow/immune cause.

18. HBV flag

WHAT YOU SEE

'HEPATITIS B FLAG'

WHAT IT ENCODES

Always screen for hepatitis B (HBsAg, anti-HBc, anti-HBs) and hepatitis C (anti-HCV) as treatable, fibrosis-driving causes in any chronic liver/fibrosis workup. Chronic HBV independently raises HCC risk (even without cirrhosis), mandating surveillance and antiviral therapy (entecavir/tenofovir) when indicated.

BOARD TRAP

WRONG answer: omitting viral hepatitis serologies and attributing all fibrosis to NAFLD/alcohol. Discriminator: HBV/HCV are reversible drivers — missing them forgoes curative DAA therapy (HCV) or HCC surveillance and antivirals (HBV); chronic HBV needs HCC screening even without cirrhosis.

19. MRCP / biopsy crossed

WHAT YOU SEE

MRCP screen and liver biopsy needle crossed out, 'ONLY IF CBC ABNORMAL'

WHAT IT ENCODES

MRCP and liver biopsy are NOT part of the routine evaluation of isolated unconjugated hyperbilirubinemia (Gilbert pattern) — they are reserved for unexplained, abnormal, or discordant cases (e.g., abnormal CBC suggesting hemolysis, cholestatic biochemistry, or noninvasive tests that disagree).

BOARD TRAP

WRONG answer: ordering MRCP or biopsy for benign Gilbert-pattern isolated indirect hyperbilirubinemia. Discriminator: with normal CBC, no hemolysis and normal ALP/transaminases the answer is reassurance — invasive/advanced imaging is over-testing; escalate only when something is genuinely abnormal.